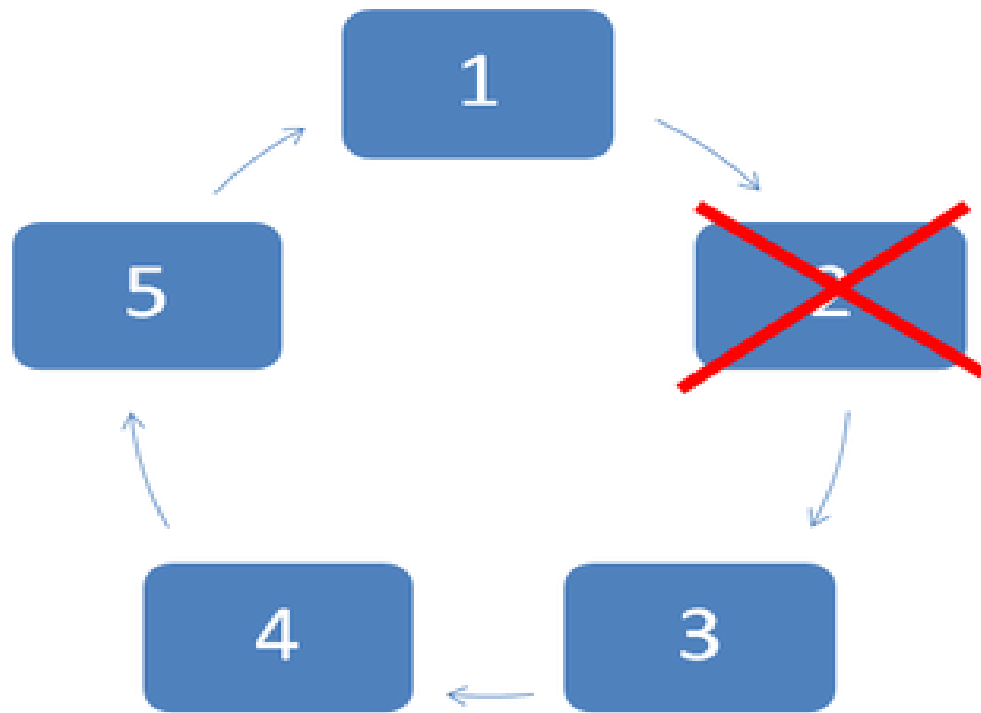




Josephus trap

- In computer science and mathematics, the Josephus Problem (or Josephus permutation) is a theoretical problem.
- **The problem statement:**
- There are n people standing in a circle waiting to be executed. The counting out begins at some point in the circle and proceeds around the circle in a fixed direction.
- **How is the Josephus problem solved?**
- The Josephus problem can be solved **using recursion**. For each iteration, recursively delete the K th position until only one person is left.

Example Diagram



Start = 0

K = 1

Size = 5

$\text{Start} = (\text{start} + k) \% \text{size}$
 $= (0 + 1) \% 5$
 $= 1$

The person at 1-index will be killed
(follow 0-indexing)

```
import java.io.*;

class Main {
    static int josephus(int n, int k){
        if (n == 1)
            return 1;
        else
            return (josephus(n - 1, k) + k - 1) % n + 1;
    }

    public static void main(String[] args){
        int n = 14;
        int k = 2;
        System.out.println("The chosen place is "
            + josephus(n, k));
    }
}
```

Output

The chosen place is 13



THANK YOU